

Agentic Financial Advisor Bot for Investment Strategies

https://www.coursera.org/learn/uol-cm3070-computer-science-final-project/peer/cVbqR/pitching-your-idea/review/r4uGhjG6EfCigwr_2iNHWQ

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1. Introduction

This proposal outlines the design and development of an Agentic Financial Advisor Bot, built to support active portfolio management through a combination of real-time financial data processing, predictive modeling, and natural language interfaces. The system is designed to be adaptable, explainable, and lightweight, providing accessible investment recommendations with a clear rationale and transparency.

2. Project Framework

The project is based on the “Financial Advisor Bot” template. The main objective is to perform active portfolio management. To meet this goal, the system integrates:

- **Data Sources:** Live financial data from Yahoo Finance API, Bloomberg API, and CCXT.
 - **AI Models:** Predictive analytics via ARIMA and reservoir computing.
 - **Reinforcement Learning Environment:** FinRL for strategic decision-making.
 - **Language Models:** Qwen3 for generating human-readable investment advice.
 - **Evaluation:** Backtesting for historical comparison and accuracy assessment.
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3. Motivation

The financial decision-making landscape is characterized by several key user pain points:

- **Information Overload:** Users face an overwhelming volume of market data.
- **Expensive Advisory Services:** Traditional financial advisors are costly.
- **Limitations of Existing Tools:** Many tools are passive and lack personalization.

Users are seeking investment tools that are simple to use, offer clear transparency, and can adapt to changing market conditions. This project aims to address those needs with an intelligent, user-friendly agentic system.

4. Related Work

Several financial platforms and research tools were reviewed to position the current work:

Project	Strengths	Weaknesses
Wealthfront [1]	Automated investing	No machine learning integration
FinRL [2]	Open-source, reinforcement learning	High technical barrier to entry
Finance Wizard [3]	Uses LLM for predictions	Functions as a black box
3Commas [4]	Visual builder for rule-based bots	Limited to predefined strategies

The current project aims to integrate the strengths of these platforms—automation, intelligence, and accessibility—while addressing their limitations, especially in transparency and usability.

5. System Design

The bot follows a structured agentic design composed of five core components:

1. **Perception:** Financial data is fetched and preprocessed using APIs like Yahoo Finance and Bloomberg.
2. **Reasoning:** Predictive models such as ARIMA and reservoir computing generate future market trends.
3. **Action Selection:** Decisions are made using FinRL to determine optimal advisory output.
4. **Natural Language Reporting:** Outputs are converted into clear, personalized advice using Qwen3.
5. **Self-Evaluation:** System performance is evaluated through backtesting, comparing predictions against actual outcomes.

An example of the system's output might be:

"Alright, after pulling the freshest data:

Current AAPL price is \$209.28 (April 25th close).

After-hours it's sitting around \$208.60.

52-week range: \$169.11 to \$260.10.

Analyst 1-year target estimate: \$236.23.

 *My Price Prediction for AAPL:*

Within the next 3 months, I predict AAPL will rise to approximately \$226.45 barring major market disruptions.

Disclaimer: This prediction is for informational purposes only."

6. Algorithm and Technical Rationale

The algorithms and tools used were selected based on a tradeoff between performance and computational efficiency:

- **ARIMA:** A lightweight statistical model suitable for short-term forecasting.
- **Reservoir Computing:** Explored for improved time-series accuracy over LSTM models [5].
- **FinRL:** Provides a simulated environment for decision-making using reinforcement learning [2].
- **Smolagents:** Used as a lightweight framework for constructing agentic systems [6].

This stack allows the system to perform real-time predictions, communicate them clearly, and improve over time through logging and evaluation.

7. Conclusion and Future Vision

The final system is:

- **Explainable:** Predictions and decisions are accompanied by natural language justifications.
- **Adaptable:** Can handle dynamic inputs and varied market conditions.
- **Lightweight:** Designed for local deployment
- **Expandable:** Designed for potential deployment via web or messaging platforms.

This bot will serve as a proof of concept for accessible, agentic financial advice leveraging AI and LLMs for clarity and trustworthiness.

References

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